Surface water salinity measurements at Son Doc, Ben Tre province, Mekong Delta, Vietnam, and monthly discharge data of Tan Chau, Mekong Delta, Vietnam

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2. Citation

When using the data please cite:

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4. Data Description

The dataset provides the non-publicly available data used in the Logistic Regression developed for predicting the salinity intrusion in the Mekong Delta. It contains measurements of surface water salinity at station Son doc, Ben Tre province in the Mekong Delta and the monthly discharge measurements at gauge Tan Chau in the Mekong Delta. The data was collected/provided by the Southern Regional Hydrometeorological Center SRHMC in Ho Chi Minh city, Vietnam and by local authorities.

The salinity measurements were taken in the surface water of the Mekong at Son Doc, Ben Tre province. The measurements cover the time span 1996 - 2016, but are discontinuous and cover only the dry (boreal winter) season. Measurements were typically taken a few days in a row in two-hour intervals, with several days without measurements in between. The discharge measurements were provided by the Southern Regional Hydrometeorological Center SRHMC in Ho Chi Minh city, Vietnam.

The monthly discharge figures were derived by SRHMC obtained from the official national hydrological gauge at Tan Chau, Vietnam, located in the Mekong (Tien in Vietnamese) branch of the Mekong river close to the border to Cambodia. Monthly discharge data were derived by SRHMC from hourly discharge measurements for the period 1995 - 2016. Data prior to 1995 was derived from historical water level measurements by the WUP-JICA project. The source of the reconstruction is provided in the data table.

5. File description

The data are provided in two .csv tables:

1. **2024-004_Apel-et-al_salinity_SonDoc_1996-2016**: Sporadic measurements of surface salinity in the Mekong river at Son Doc, Ben Tre province, Mekong Delta, Vietnam (latitude: 10.06425° N, longitude: 106.45747° E)
2. **2024-004_Apel-et-al_discharge_Qmonthly_TanChau_1964-2017_gaps-filled**: Monthly mean discharge at gauge Tan Chau, Mekong (Tien) river, Mekong Delta (latitude: 10.80065° N, longitude: 105.24798° E)

Table *1*. *2024-004_Apel-et-al_salinity_SonDoc_1996-2016.csv* contains the surface salinity measurements at Son Doc (Ben Tre province, Mekong Delta, Vietnam). Column **DATETIME** contains the date and time stamp of the measurements, column **SALINITY [g/l]** contains the salinity readings. Furthermore, information about the location and coordinates in decimal degrees are provided.

Table *2*. *2024-004_Apel-et-al_discharge_Qmonthly_TanChau_1964-2017_gaps-filled.csv* contains the monthly discharge measurements in Tan Chau (Mekong Delta, Vietnam). Column **year** contains the year of data point, columns headed by 01-12 indicate the **month** of the year. The monthly discharge values are provided in **Q [m/s]**.

NOTE: Data from 1995 until 2016 are from the official gauging station. 1980-1995 only water level, no discharge measurements are available. The data in the table prior 1995 was reconstructed from daily water level to discharge relations by the WUP-JICA project. A report is available at: http://open_jicareport.jica.go.jp/pdf/11755576_03.pdf (WUP-JICA 2004). The reconstructed time series can be found on page 42 of that report.

6. References

Apel, H., Khiem, M., Quan, N.H., and Toan, T.Q. (2020). Brief communication: Seasonal prediction of salinity intrusion in the Mekong Delta. Nat. Hazards Earth Syst. Sci. 20(6), 1609-1616. https://doi.org/10.5194/nhess-20-1609-2020

WUP-JICA (2004) Hydrological Conditions in the Mekong Delta. Supporting Report, Paper VII: Maintenance of Flows on the Mekong Mainstream. URL: http://open_jicareport.jica.go.jp/pdf/11755576_03.pdf